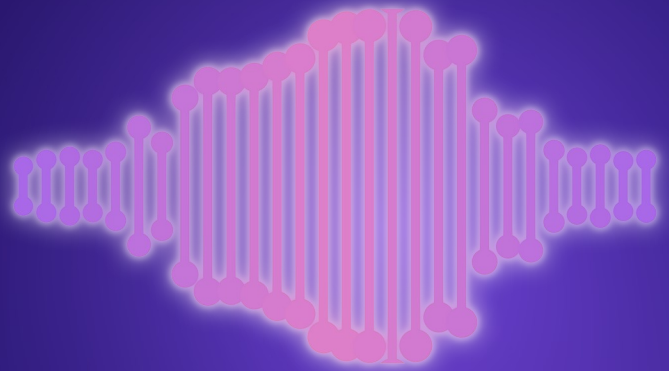
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NURSE STRESS STUDY | WEARABLE DATA

# Does a wristband know how a nurse feels?



Wearable data from 15 hospital nurses was matched to the stress levels they reported.

**15** nurses, about 1,250 hours of wristband data

**214** reported work periods with sensor data

**4** body signals: skin conductance, heart rate, skin temperature, movement

1. Line up each report with the wristband data
2. Compare high-stress periods with the rest
3. Allow for each nurse's own normal

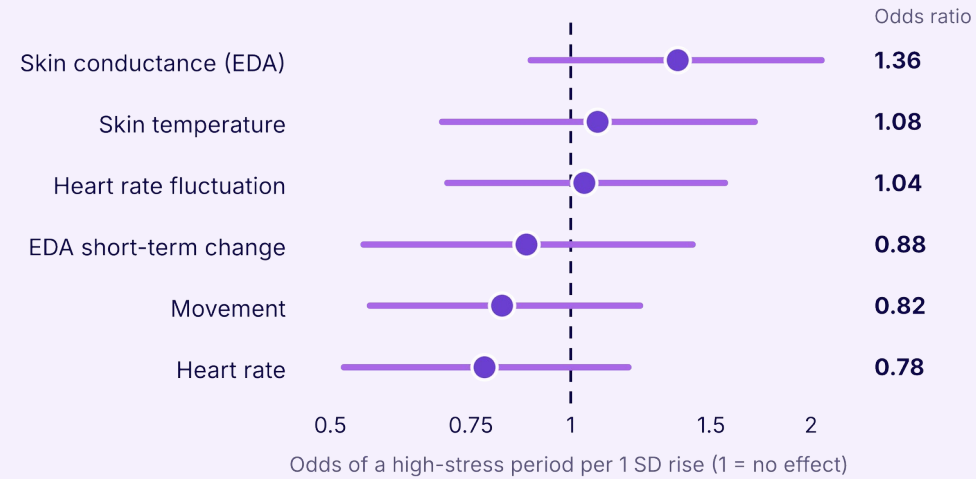
Source: Nurse Stress Dataset (Empatica E4), Dryad doi:10.5061/dryad.5hqbzkh6f

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WHAT THE DATA SHOW

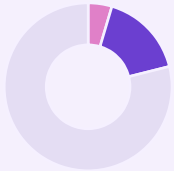
# Wrist signals alone did not flag stress

## Does any signal predict a high-stress period?




*Every bar crosses 1, so no signal clearly raises or lowers the odds.*

## What explains a high-stress period?



- **4.6%** the 6 body measures
- **16.5%** which nurse it is
- **78.9%** not explained

- No signal separates high-stress periods from the rest.
- Who the nurse is matters more than the signals.
- 73% of reports were the top level: few calm periods to compare.

Source: Nurse Stress Dataset (Empatica E4), Dryad doi:10.5061/dryad.5hqbzkh6f

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WHAT IT MEANS

# Wearables show the body. People report how it feels.

## What it means

- A wrist band on its own cannot say who is stressed right now.
- Each nurse has their own normal, so one threshold for everyone may misfire.

## Limitations

- Only 214 reported periods; about 8% of time is labelled.
- Stress is reported once per period.
- Associations only; the level 2 cutoff is an analyst choice.

## Next steps

- Compare each nurse against their own baseline.
- Test stressor types, such as patient crisis or workload.
- Validate on new nurses with regular check-ins.

**Next: personal baselines plus patient check-ins →**

Source: Nurse Stress Dataset (Empatica E4), Dryad doi:10.5061/dryad.5hqbzkh6f

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HOW IT COMPARES

# The analysis lines up with the dataset paper

| Signal | What the paper reports | What the analysis found | Match |
|---|---|---|---|
| **Skin conductance (EDA)** | Most informative signal in its stress detector; a positive link for some nurses, not all | Largest odds ratio (1.36), but the range includes 1 | Partly |
| **Heart rate** | No strong link to stress; also driven by physical activity | No link (odds ratio 0.78, correlation -0.03) | Consistent |
| **Skin temperature** | Higher with high stress, but not significant (p = 0.22) | Weak positive link (correlation 0.13), not significant | Consistent |
| **Nurse differences** | Signals vary between nurses; personalised models advised | Nurse explains about 17% of the picture, signals about 5% | Consistent |
| **Overall** | Wearables complement, not replace, reports and interviews | A wristband alone did not flag stress | Consistent |

**Where the analysis differs:** the paper's random forest predicted stress from short signal windows, and its labelled periods were first flagged by that detector. All signals were tested together, adjusted for nurse, so the numbers are not directly comparable.

Sources: Dryad dataset doi:10.5061/dryad.5hqbzkh6f; Hosseini et al., Scientific Data 9:255 (2022), doi:10.1038/s41597-022-01361-y